

Total No. of Questions : 10]

SEAT No. :

P-2965

[Total No. of Pages : 3

[6004]-948

**B.E. (Mechanical/Mechanical SW)  
ENERGY ENGINEERING  
(2015 Pattern) (Semester - II) (402047)**

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right of each question indicate full marks.
- 4) Assume suitable data wherever necessary and mention the same clearly.
- 5) Use of steam tables. Mollier chart and calculator is allowed.

- Q1)** a) What do you mean by cascade efficiency? Explain in brief. [5]
- b) What are the sources of air leakage in condenser and Explain its effects on condenser performance. [5]

OR

- Q2)** a) Draw layout of pulverized fuel handling systems. List its advantages and disadvantage. [5]
- b) What are the different pollutants from thermal power plants? Explain their effects on human health. [5]

- Q3)** a) Write a short note on hydrograph and unit hydrograph. [5]
- b) Explain with a neat diagram the process of coal beneficiation. [5]

OR

- Q4)** a) What do you mean by cladding in nuclear power plant? Which materials are used for cladding? What are the purposes of cladding? [5]
- b) Explain the necessity of steam condenser in steam power plants. [5]

P.T.O.

- Q5) a)** An open cycle gas turbine power plant works on Brayton cycle. The maximum pressure and temperature of the cycle are limited to 5 bar and 900K. The pressure and temperature of the gas entering into the compressor are 1 bar and 27°C. Reheating is used at a pressure of 2.5 bar where the temperature of the gases is increased to its original turbine inlet temperature. The air flow per second through the plant is 10 kg/sec. Determine the thermal efficiency and plant capacity in MW. The exhaust pressure of the turbine is also 1 bar.

Assume the compression and expansion is isentropic.

Take  $\gamma = 1.4$  for air and gases,  $C_P = 1 \text{ kJ/kg-K}$  for air and gases  
 $C.V$  of fuel = 40,000 kJ/kg.

Neglect the pressure losses in the system. Do not neglect fuel quantity. [8]

- b) Draw diesel power plant layout and explain the working. [8]

OR

- Q6) a)** Discuss the effect of Gas Turbine Cycle with [8]

i) Regeneration

ii) Intercooling

- b) Discuss following for Diesel power plant [8]

i) Fuel and Fuel Injection System

ii) Engine Cooling

iii) Lubricating System

- Q7) a)** Write short note on Flash steam Geothermal power plant. [6]

- b) Discuss the working principle of solar pond based power plant with diagram. [6]

- c) Discuss the environmental impacts of non-conventional power plant. [6]

OR

- Q8) a)** Elaborate working of Fuel cell. What are the different types of fuel cells? [6]

- b) Explain classification of wind turbines. Elaborate operating characteristics of Wind Power Plant. [6]

- c) Write short note on Close cycle MHD generators. [6]

**Q9) a)** State the various protective equipment and explain the working of switchgear in power plant. [8]

b) A 60 MW power station has an annual peak load of 50MW. The power station supplies loads having maximum demand of 20 MW, 17MW, 10MW and 9MW. The annual load factor is 0.45. [8]

Find :

- i) Average load
- ii) Energy supplied per year
- iii) Diversity factor
- iv) Demand factor

OR

**Q10)a)** Write short notes on : [8]

- i) Circuit breaker
- ii) Power transformers

b) Define and write significance of the following terms : [8]

- i) Load duration curve
- ii) Demand factor
- iii) Load factor
- iv) Group diversity factor

